How Climate Change Affects the World We Live In

A Report by Choolwe Cheelo

Overview

Climate change is a cause that has received more attention in recent years. You might have wondered what all the worry has been about or even fallen in the trap of thinking it is just a hoax created to scare people. But that is not the case, climate change is real and we are already seeing the world around us change because of it. We are going to look at a few of the indicators and causes of climate change and the effect it has had on our population.

Climate change is a long-term change in the average weather patterns that have come to define Earth's climate. Changes observed in Earth's climate since the mid-20th century are driven by human activities, particularly fossil fuel burning, which increases heat-trapping greenhouse gas levels in Earth's atmosphere, raising Earth's average surface temperature.

Scientists use observations from the ground, air, and space, along with computer models, to monitor and study past, present, and future climate change. Climate data records provide evidence of climate change key indicators, such as global land and ocean temperature increases; rising sea levels; ice loss at Earth's poles and in mountain glaciers; frequency and severity changes in extreme weather such as hurricanes, heatwaves, wildfires, droughts, floods, and precipitation changes.

What Our Data Tells Us

An analysis of three of the most crucial factors was done to determine their effect on climate change. CO2 emissions, temperature, and forest area were analysed by region and increase over the years.

Fossil fuels – coal, oil and gas – are by far the largest contributor to global climate change, accounting for around 68 per cent of global greenhouse gas emissions and 90 per cent of all carbon dioxide emissions.

As greenhouse gas emissions blanket the Earth, they trap the sun's heat. Because of this, the planet is warming up and the climate is changing. The world is now warming faster than at any point in recorded history. Warmer temperatures over time are changing weather patterns and disrupting the usual balance of nature. This poses many risks to human beings and all other forms of life on Earth.

Forests cover 33 per cent of the Earth's land surface. They provide habitat to some 80 per cent of land-based species. They filter water, recycle rainfall, and anchor the soil, and directly support the livelihoods and well-being of about 25 per cent of humanity.

Forests are also one of the planet's most powerful tools against climate change. Through photosynthesis, trees remove climate-warming carbon dioxide from the atmosphere and release oxygen. By locking carbon pollution in trees, roots and soil, forests reduce the heat-trapping gases and regulate Earth's temperature.

In the last decade alone, forests, plants and soil absorbed around 30 per cent of all carbon emissions generated by the burning of fossil fuels and other human activities.

The cutting down of trees is putting forests, and everything they provide, at risk. At the same time, more extreme heat, and droughts – due to climate change – are increasing the likelihood of forest fires. Fires accounted for almost half (44 per cent) of all tree cover loss between 2023 and 2024. When trees are felled or burnt and decay, they release the carbon they have stored. The world's forests risk shifting from a carbon sink to a carbon source.

Climate Change Impact on Earth

As climate change intensifies, there is no question that the intensity and frequency of extreme weather—often resulting in disasters—is increasing. According to the IPCC's most recent report on climate adaptation, disasters fueled by climate change are already worse than scientists originally predicted.

In some landscapes, flooding can be a natural part of a yearly cycle, providing ecosystem services and supporting livelihoods yet excessive flooding has impacted so many lives. Heavy rainstorms are projected to become more common and more intense due to higher temperatures, with flash floods expected to become more frequent across the globe. In fact, floods impact more people worldwide than any other disaster, and the economic, social, and environmental impacts are getting worse.

With climate change intensifying, hotter temperatures, more intense and longer dry seasons, earlier snowmelt, and stronger winds damage nature's ability to resist fire. A drought is an unusual and temporary deficit in water availability caused by the combination of lack of precipitation and more evaporation (due to high temperatures). More frequent and severe droughts will increase the length and severity of the wildfire season.

Climate change is increasing ocean and atmospheric temperatures and causing higher sea levels, which in turn can increase the frequency, duration, and intensity of hurricanes, along with their peak winds, storm surge, and rainfall rates.

What Can We Do?

Climate change is happening now, and it is the most serious threat to life on our planet. Since 1980, 7.8 billion people have been affected by climate related disasters in one way or another. And the frequency of these disasters has been increasing steadily over the years. There is a 98% likelihood that at least one of the next five years, and the five-year period, will be the warmest on record.

The longer countries wait to respond, the greater the damage and cost to contain it. One estimate from Columbia University put the cost of adaptation needed just for urban areas at between US$64 billion and $80 billion a year – and the cost of doing nothing at 10 times that level by mid-century.

Risk associated with weather-related hazards is disproportionately concentrated in developing countries and within these countries in poorer sectors of the population. Poverty and constrained access to productive assets mean that rural livelihoods that depend on agriculture and other natural resources are vulnerable to even slight variations in weather and seasonality.

In 2015, world leaders signed a major treaty called the Paris agreement to put these solutions into practice. Core to all climate change solutions is reducing greenhouse gas emissions. Because both forests and oceans play vitally important roles in regulating our climate, increasing the natural ability of forests and oceans to absorb carbon dioxide can also help stop global warming.

The main ways to stop climate change are to pressure government and business to keep fossil fuels in the ground, invest in renewable energy, switch to sustainable transport, help us keep our homes cosy, improve farming and encourage vegan diets, restore nature to absorb more carbon, protect forests, protect the oceans, and reduce plastic.

If we do these things, there just might be a chance to stop the damage from increasing any further.

Conclusion

The findings suggest that continued increases in emissions may lead to further temperature rises and intensified climate-related impacts. Overall, the analysis highlights that climate change is a diverse issue driven by emissions, influenced by environmental factors such as forest coverage, and reflected in significant human impacts. Addressing these challenges requires coordinated efforts across environmental, economic, and policy dimensions.

It might seem like it is too late to do anything about climate change, but the effects and disasters cannot be ignored. If we all do our part to reduce global emissions, plant more trees, and stop global warming, we could see a reduction of the devasting climate related disasters we face.